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1. Hospital patient segmentation

K	SSE
1	600
2	350
3	220
4	200
5	190

optimal number of clusters = 3

These 3 clusters can keep the hospital identify different patient group.

→ High - risk patiens - require closes monitoring and regular treatment.

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→ Medium - risk patients - require periodic health care support

→ low - risk patients - require routine health care servical

string 3 clusters provides meaningful segmentation. without creating to many groups.

E - commercl order analysis :-

In hierarchical clustering, orders are grouped based on their similarity in product

From the dataset

→ Elements :- 01 = 55,000 and 03 : 57,000 → very similar.

→ Clothing :- 02 = 25,000 and 05 = 27,000 → very similar

→ furniture :- 04 = 30000 → forms a seperate group intially.

These clusters help the retailers understand purchasing patterns, high value electronics customers may be targeted.

Mobile App uses Behaviour

In soft clustering a user can belong to more than one cluster with different membership probabilities.

U1 : High usage and frequent logen → mainly highly active

U2 : low usage but High Purchase → may have membership in bote,

U3. very High usage and login frequency.

U4: low usage and low login frequency.

U5: High usage and frequent login.

(5). Employee performel evaluation

model	SS
K = 2	0.48
K = 3	0.68
K = 4	0.52

The highest silhoutta score is 0.68 for $K=3$

∴ The most suitable number of clusters is 3

A possible segmentation is :

1. High - performing employees High productivity attendence and skills.

2. medium - performing employee - moderate score.

3. low performing employees - lower productivity attendence and still score

• A silhoust score closes to 1 indicates bets defined and weal - soporated clusters since $K=3$ has the highest score.

4. Agricultural crop monitoring

Before normalization.

- Rainfal : 280 - 520mm
- Soil fertility : 3 - 9
- crop yeid : 65 - 90 tons

Because rainfall has much larges numerical values it can dominate distances calculation in clustering.

Effect of normalization :

1. prevents large - scale features from dominating clustering
2. makes distance calculations fair
3. produces more meaning ful clusters
4. Improves comparison b/w forms
5. makes agricultural analysis more reliable.